GCS North Star Metric

Account Value Realization Index (AVRI)

> *"PM is a translation function, between business intent, data reality, and shippable software."*

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GCS is moving from selling deals to sustaining value.

TCV-based compensation rewarded the signature. The new ARR + consumption hybrid only pays out if customers actually use what they bought. We need ONE metric that balances four very different things.

OLD WORLD: TCV

Book the deal. Bank the credit.

One signal: the contract was signed.
One moment: the day it closed.
One incentive: land it big and walk away.

Blind to: deployment · health · sustained usage

NEW WORLD: ARR + CONSUMPTION

Bookings

Deployment

Tech Health

Sustained Usage

Customer pays only if value is realized. Compensation must reflect the same.

Need: ONE metric that balances all four

The business question I'm answering: ***"What's the health of an account at a point in time, balancing all four dimensions, in a single defensible number?"***

Every existing metric is blind to ≥1 of the four dimensions.

I scored 30+ industry-standard SaaS/CS metrics against the four required dimensions. None covers all four. The closest pattern, Gainsight-style composites, has known gaps for consumption-hybrid businesses.

Metric	Bkgs	Deploy	Tech	Usage	Lead/Lag
TCV (Total Contract Value)	●	○	○	○	Lag
ARR (Annual Recurring Rev)	●	○	○	○	Lag
NRR (Net Revenue Retention)	●	○	○	◐	Lag
Raw Utilization (90d)	○	◐	○	●	Lag
Latest Health Color	○	○	◐	○	Lead
Gainsight CHS (composite)	◐	◐	●	●	Lead
AVRI (proposed)	●	●	●	●	**Lead**

● covers ◐ partial ○ blind

Build vs Buy: *Gainsight CHS is the closest commercial alternative, same composite pattern, but doesn't natively handle commit-vs-consume, trajectory, or the 5 brief-specified edge cases. AVRI is that pattern, specialized for PANW's consumption-hybrid model.*

AVRI × naive CHS, 730 in-scope accts

	CHS GREEN	CHS YELLOW	CHS RED
AVRI GREEN	444 $181.5M ARR	89 $33.5M ARR	—
AVRI YELLOW	7 $400K ARR	48 $20.2M ARR	8 $4.9M ARR
AVRI RED	—	25 $16.9M ARR	109 $47.7M ARR

Diagonal (agree): 601 accts. Off-diag (disagree): 129 accts. Plus 36 onboarding (v1.3 grace, deferred).

AVRI surfaces 22% Red vs CHS's 19%, AND $22M ARR sits where AVRI flags risk that naive CHS misses.

AVRI: the quality term — 0–100 composite that balances all four dimensions.

CR Commit Realization

weight 30%

Are they using what they paid for?

90-day consumption ÷ 90-day commit. Piecewise: penalize <50%, plateau 50–110%, gentle decay >150%.

UM Usage Momentum

weight 30%

Is consumption holding up?

90-day avg ÷ 12-month baseline. Catches spike-and-drop. Level guard zeros UM if annual usage <5% of commit.

DM Deployment Maturity

weight 20%

Is the product genuinely operationalized?

Active days in last 90 ÷ 90. Production version: also license provisioning, feature breadth, milestones.

TH Technical Health

weight 20%

Does it work for them?

Exponentially-decayed health color (0.95/day). Production: also Sev-1 freq, SLA attainment, escalations.

$$AVRI = 0.30 \cdot CR + 0.30 \cdot UM + 0.20 \cdot DM + 0.20 \cdot TH$$

Floor rule: if TH < 30, cap AVRI at 50.

GREEN ≥ 75
Healthy · expand

YELLOW 50–74
Watch · CSM action

RED < 50
At risk · escalate

v1.3 — Activation Grace: *newly-signed contracts excluded from CR until cumulative usage ≥ 15% of monthly commit OR 90 days elapsed (whichever first). Signing a deal never penalizes a CSM during ramp.*

Realized Value: AVRI is velocity. RV is momentum. Both belong on the dashboard.

AVRI scores quality of execution; segment-fair and comp-grade. RV composes ARR with AVRI to surface dollars at stake; the executive-dashboard metric. Three uses, three consumers, one engine.

THE FORMULA

$$RV = ARR \times AVRI / 100$$

Linear v1. Decomposable. Calibratable. Onboarding accounts contribute zero.

THE HEADLINE

$291M ARR • Realizing 78% • $65M unrealized • 40 accounts in grace

Decomposes by pillar.
Σ of CR + UM + DM + TH + Floor losses = total unrealized $.

Sums at every level.
Account → CSM → Region → Org. Numbers reconcile.

UNREALIZED $ BY PILLAR (org-wide)

AVRI — QUALITY OF EXECUTION

Segment-fair, comp-grade. Used for CSM evaluation, performance review, and the case-study disagreements with naive CHS.

RV — VALUE AT STAKE

Scale-aware. Used for executive risk concentration: "where are dollars leaking?" Drives the dashboard's headline and pillar heatmap.

UNREALIZED $ — OPERATIONAL TRIAGE

Account-level. Used by individual CSMs to triage their book: "which 5 accounts have the most $ at risk this week?"

Why linear, not quadratic? Decomposability + honesty (no curvature without renewal data) + calibratability for v3.

Three disagreements with naive CHS, each validating a different design choice.

Three representative cells from the AVRI × CHS matrix on slide 3. Each off-diagonal disagreement validates a specific AVRI mechanism: floor rule, decay-weighted TH, and consumption-dominant weighting.

FLOOR RULE FIRES

ACC-00026

Healthcare · $165K ARR · 130 days

CHS 78 (green)

AVRI 50 (yellow)

Cell: AVRI Yellow × CHS Green
RV $83K · Unrealized $83K (50% of book on this account)

All three consumption pillars at 100, but TH = 28.5. Floor rule caps AVRI at 50 — refuses to let usage paper over a tech-health crisis. Without the rule, AVRI = 86 (Green). CHS has no analog.

DECAY WEIGHTING

ACC-00876

Manufacturing · $246K ARR · 215 days

CHS 57 (yellow)

AVRI 92 (green)

Cell: AVRI Green × CHS Yellow
RV $227K · Unrealized $19K (the easy-win contrast)

All pillars high; latest health color happened to be red (one bad week). CHS over-penalizes recency. AVRI's 90-day decay-weighted TH (≈61) sees the trend, not a single data point.

AVRI ESCALATES

ACC-00298

Financial Services · $325K ARR · 332 days

CHS 57 (yellow)

AVRI 24 (red)

Cell: AVRI Red × CHS Yellow · most-common disagreement
RV $77K · Unrealized $248K (the renewal landmine)

5% util, no momentum, sparse engagement — three pillars in collapse. CHS sees ARR + recent green color and says 'watch.' AVRI escalates. 25 accounts / $16.9M ARR sit in this cell.

→ **LIVE DEMO** Streamlit · At-Risk Renewals tab · ***$10.3M across 20 accounts renewing in 90 days, flagged Yellow or Red.***

HUMAN IN THE LOOP *AVRI scores; dashboard surfaces; CSM acts. The metric directs the human, not replaces them.*

SUCCESS CRITERIA (1) % of Red accounts churning in 90d (2) Reduction in Green→Lost surprise renewals (3) Expansion conversion in 110–150% util band

What I learned, what v2 looks like, how to ship it.

Built end-to-end in ~8 hours of focused spec-driven AI work: data generator (Python), BigQuery pipeline (SQL), dashboard (Streamlit), 13 documented pitfalls captured in lessons_learned.md. Below: what changes next.

01 What worked + what I'd change

- Specs first, AI codes, the markdown specs are what survive.
- Inductive workflow (data → metrics → refine) beats deductive, stronger narrative for the panel.
- Pre-write validation gates caught structural problems before BigQuery.
- Next time: build CI from day one, would've caught UM zero-baseline bug 2 hours earlier.
- Separate LLMs for code-gen vs code-review.

02 AVRI v2, newer tech

- **v2 SHIPPED: Realized Value (linear), pillar decomposition, calibration sandbox, 29 DQ tests pass.**

Agentic DQ: agent watches the pipeline, surfaces anomalies in Slack autonomously.
LLM-summarized account briefs: per at-risk acct, generate CSM-ready talking points from history.
Per-segment AVRI weight tuning (Enterprise vs Mid-Market), calibrated against renewal correlation.
Sigmoidal RV: refit AVRI→realized-revenue function from historical renewals.
Replace TH proxies with real signal: Sev-1 frequency, SLA attainment, escalations.

03 30/60/90 day rollout

- Day 30, Validate AVRI against historical renewal outcomes; calibrate weights via regression.
- Day 60, Integrate ticket data + license provisioning; replace proxy signals.
- Day 90, Roll into CSM compensation as a 25% factor (alongside book ARR retention).
- Day 180, Dashboard goes from prototype to production (Looker/Tableau for execs; Streamlit retained for diagnostics).
- Day 365, Re-tune weights based on first full year of data.

The repo is reproducible end-to-end. Anyone on your team can clone it and have AVRI scoring real data on day one.

How AVRI handles the brief's five edge cases.

Each edge case is answered at the right altitude — pillar mechanism, pipeline logic, or DQ layer. Naive CHS treats every problem as a scoring problem; AVRI separates structural design from data-quality.

EDGE CASE	NAIVE CHS HANDLING	AVRI MECHANISM	VERDICT
Spike & Drop ~5%	Recent health color still green from burst period. ARR component still high. Hedges to Yellow.	UM level guard fires when 90d/annual usage collapses. CR window-based (90d, not annual) catches the drop.	RED UM · CR
Shelfware ~10%	ARR component (often 30%+ weight) props score into the 30–50 range. Looks like a 'watch' account, not an emergency.	ARR not a pillar. Three pillars collapse (CR≈0, UM≈0, DM≈0). Even perfect TH pulls AVRI to ~20.	DEEP RED CR · UM · DM
Consistent Overage ~15%	Caps util at 100% or penalizes anything above. 130%-util account scores Yellow; CSM told to 'manage burn rate.'	CR is piecewise: 50–110% util = full credit (sweet zone). 110–150% = gentle decay. 200%+ = hard zero.	GREEN + expansion CR plateau
Mid-Year Expansion ~5%	Takes latest contract only (loses original commit) OR sums both regardless of date overlap (double-counts).	Denominator at any point in time = SUM of active commits for any contract whose date range covers that point.	no distortion Pipeline
Orphan / Rogue Usage ~300 logs	LEFT JOIN shortcut: orphan logs vanish silently; out-of-window logs credited as billable. Rollups inflate.	Pipeline INNER JOINs daily_usage_logs to contracts on date range. Out-of-scope rows excluded from CR/UM.	QUARANTINED DQ layer

3 of 5 edge cases caught by structural pillar design · 1 by pipeline logic · 1 by data-quality layer. Different problems, answered at the right altitude.